

DEEP LEARNING

Open-Book Test

Duration: 60 Minutes

Maximum Marks: 20

You are debugging an ultra-deep neural network that is failing to learn. The network uses

$$W_{ij} = 5.0$$

and the Hard Sigmoid activation

$$\sigma(z) = \begin{cases} 0, & z \leq -2.5, \\ 0.2z + 0.5, & -2.5 < z < 2.5, \\ 1, & z \geq 2.5. \end{cases}$$

It is trained using Adam with

$$\beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon = 10^{-8}, \quad \eta = 10^{-3}.$$

Assume that the initialization places the representative neurons in the saturated regime

$$z \geq 2.5,$$

so that

$$\sigma(z) = 1, \quad \sigma'(z) = 0.$$

For the independent Adam analysis, assume a representative parameter receives a constant nonzero gradient

$$g_t = 10^{-12}, \quad t = 1, 2, 3,$$

with

$$m_0 = v_0 = 0.$$

No specific network architecture is required. Analyze gradient propagation qualitatively; all numerical calculations should use only the quantities provided below.

1. **[8 Marks]** Qualitatively explain how the large positive initialization and Hard Sigmoid saturation can suppress gradient propagation through an ultra-deep network. Then calculate Adam's first- and second-moment estimates m_t and v_t for each of the first three steps using

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t,$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2.$$

2. [12 Marks] Calculate the bias-corrected estimates

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t},$$

the Adam scaling term

$$\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}},$$

and the corresponding parameter update

$$\Delta\theta_t = -\eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

for $t = 1, 2, 3$. Compare the resulting update with the corresponding SGD update. Finally, distinguish between a very small nonzero gradient (10^{-12}) and an exactly zero gradient caused by Hard Sigmoid saturation, and explain why Adam cannot compensate for the latter.

Note: The test is open-book. Show the essential mathematical steps and clearly distinguish the prescribed gradient used for the Adam analysis from the saturated-activation scenario.